

Al-Azhar University – Gaza
Faculty of Engineering & IT
Department of IT & CSE
Instructor: Dr. Izzeddin Alshawwa



Course Code: ITCS4316
Course Name: CNs
1st Semester – 2025/2026
Mon. 05/01/2026, 10⁰⁰ - 17⁰⁰

St. Name:

Mid Exam
(Total Marks: 40)

St. Reg. #:

ANSWER ON ALL QUESTIONS

Q1: What are two reasons for using layered protocols? What is one possible disadvantage of using layered protocols? **(5 pts).**

Q2: When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss these two approaches. **(5 pts).**

Q3: List two advantages and two disadvantages of having international standards for network protocols. **(5 pts).**

Q4: A network engineer must choose between coaxial cable and fiber optics for a campus backbone. The campus experiences strong electromagnetic interference from nearby equipment. Which medium is more suitable? Justify your answer using physical layer properties **(5 pts).**

Q5: Analyze the significance of protocol layering in network design and its role in ensuring network reliability and evolvability, particularly when integrating new technologies with existing infrastructure **(5 pts)**.

Q6: Analyze the different methods for connecting a home to the Internet, comparing DSL, cable modem, and fiber optics in terms of bandwidth, latency, and deployment costs. How do these factors influence the choice of connection type for various users? **(5 pts)**

Q7: Evaluate the challenges and benefits of integrating power-line networking with traditional Ethernet and Wi-Fi systems in a smart home environment. How do power-line networks overcome limitations in Wi-Fi coverage, and what are the potential drawbacks? **(5 pts)**

Q8: What are the advantages of fiber optics over copper as a transmission medium? Is there any downside of using fiber optics over copper? **(5 pts)**

Best Wishes